

Exercise 11

Chapter 2, Page 76

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Introduction to Electrodynamics

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It should be clear that, since the charge distribution is spherically symmetric, the field has to be radially oriented, and should only depend on r . Let $\mathcal{S}_r$ be a sphere centered around the origin. The charge lays on a sphere $\mathcal{S}_R$ and has the surface density σ .

Step 2

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Inside the $\mathcal{S}_r$, where $r < R$, there is no charge. If $r > R$, however, $\mathcal{S}_r$ encloses the entire charged sphere and the total enclosed charge is equal to $4\pi\sigma R^2$, the area of the sphere times its charge density. Therefore, according to the Gauss law, we have

$$\begin{aligned}
 \oint_{\mathcal{S}_r} \mathbf{E} \cdot d\mathbf{a} &= \oint_{\mathcal{S}_r} E da \\
 &= E(r) \oint_{\mathcal{S}_r} da \\
 &= 4\pi r^2 E(r) \\
 &= \frac{1}{\epsilon_0} Q_{\text{enc}}(\mathcal{S}_r) \implies \\
 E(r) &= \begin{cases} 0 & \text{if } r < R \\ \frac{\sigma R^2}{\epsilon_0 r^2} & \text{if } r > R \end{cases}
 \end{aligned}$$

Step 3

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The answer is the same as in problem 2.7. since $Q = 4\pi R^2\sigma$. By comparing the calculations in this problem and problem 2.7., we come to appreciate the elegance of the Gauss's law.

Result

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$$E(r) = \begin{cases} 0 & \text{if } r < R \\ \frac{\sigma}{\epsilon_0} \frac{R^2}{r^2} & \text{if } r > R \end{cases}$$

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